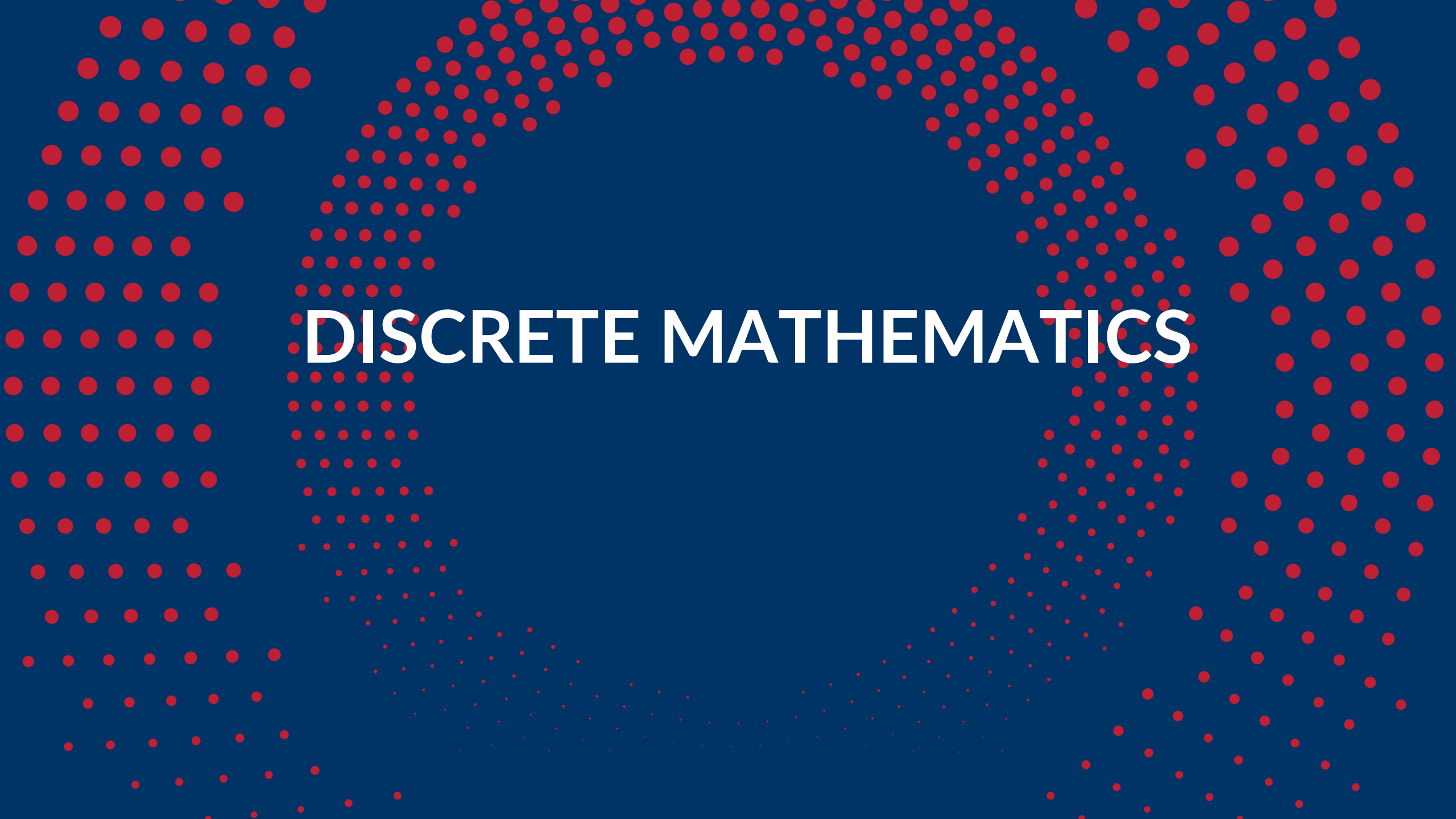




HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.



DISCRETE MATHEMATICS



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PART1: COMBINATORIAL THEORY

PART 2: GRAPH THEORY

Contents of part 1

Chapter 0: Sets, Functions

Chapter 1: Counting problem

Chapter 2: Existence problem

Chapter 3: Enumeration problem

Chapter 4: Combinatorial optimization problem



Chapter 2. Existence problem

1. Introduction to existence problems
2. Basic proof methods
3. Dirichlet principle (pigeonhole principle)

1. Introduction to existence problems

- In the “Counting problem” chapter, we focused on counting the combinatorial configurations. In those problems, the existence of the configurations is obvious, and the main object is to count the number of elements that satisfy the given properties.
- However, in many combinatorial problems, it is very difficult to point out the existence of a configuration that satisfies given properties:
 - For example, when a player needs to calculate his moves to answer whether there is a possibility of winning or not?
 - A person needs to search for the key to decipher a secret code that he does not know if this is really the opponent's encrypted message, or just the secret code issued by the opponent to ensure the safety of real telegrams ...
- In combinatorics, besides the counting problem, there is another very important problem is considering the existence of combinatorial configurations satisfying given properties - the problem of existence.

1. Introduction to existence problems

The 36 officers:

- This problem is proposed by Euler, it is described as following:

“You're in command of an army that consists of six regiments, each containing six officers of six different ranks. Can you arrange the officers in a 6x6 square so that each row and also each column of the square holds officers of all 6 ranks and all 6 regiments?”

1. Introduction to existence problems

The 36 officers:

- Using:
 - A, B, C, D, E, F refer to 6 regiments, respectively
 - 1, 2, 3, 4, 5, 6 refer to 6 ranks of officers, respectively.
- This problem could be generalized by replacing 6 by n .
- In the case $n = 4$, a solution to the problem is:

A1	B2	C3	D4
B4	A3	D2	C1
C2	D1	A4	B3
D3	C4	B1	A2

- In the case $n = 5$, a solution to the problem is:

A1	B2	C3	D4	E5
C4	D5	E1	A2	B3
E2	A3	B4	C5	D1
B5	C1	D2	E3	A4
D3	E4	A5	B1	C2

1. Introduction to existence problems

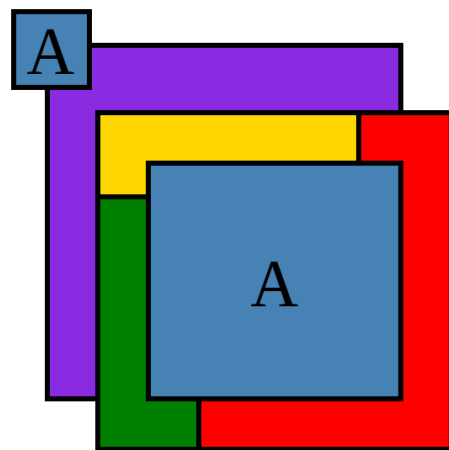
The 36 officers:

- As the solution to the problem can be represented by combining two same size squares array of symbols: one with upper-case letter and the other with numbers. The problem is also called as Graeco-Latin square (also called an Euler square)
- Euler struggled to find the solution to the 36 officers problem but was unsuccessful. However, the solution exists when $n = 4, 5$ and 7 . Thus, he proposed a general hypothesis: The Graeco-Latin squares are impossible if $n = 4k + 2$.
- Tarri, in 1901, proves the correct hypothesis for $n = 6$, by examining all possible ratings.
- In 1960, three American mathematicians, Boce, Parker, and Srikanda showed a solution with $n = 10$ and then showed a method to construct Graeco-Latin squares for all $n = 4k + 2$, with $k > 1$.

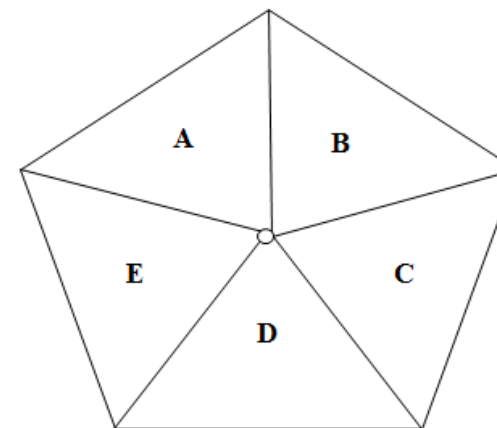
1. Introduction to existence problems

The 4-color problem:

- There are problems that its content could be explained to anyone, but its solutions anyone can try to find, but it is difficult to find. The 4-color problem is such a problem.
- The problem can be stated visually as follows: Prove that all maps on the plane could be colored with 4 colors so that no two adjacent regions (neighbor regions) are colored by the same color.
- Note that each country is considered a **connected region** and the two countries are called neighbors if they **share a border of a continuous line**.



A is not connected region

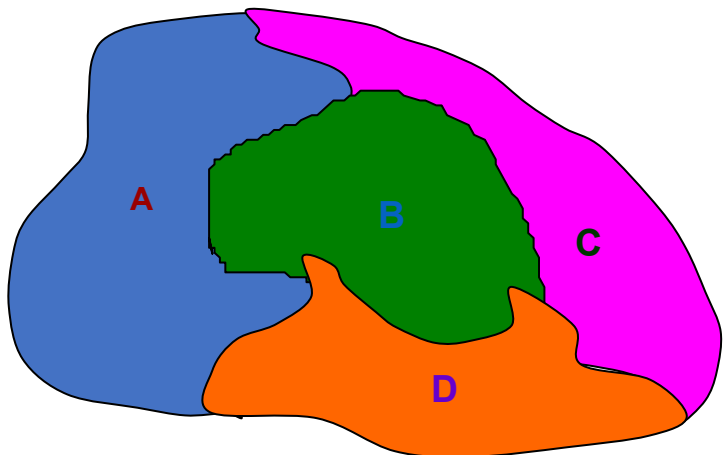


A and C are not neighbors

1. Introduction to existence problems

The 4-color problem:

- The number 4 is not random. It has been shown that all maps are colored with color numbers greater than 4, but with color numbers less than 4 it cannot be colored. For example, the map with 4 countries in the picture below cannot be colored if using less than 4 colors.



1. Introduction to existence problems

The 4-color problem:

- This problem was first proposed in 1852 by Francis Guthrie, while trying to color the map of counties of England.
- In 110 years many proofs have been published but have been flawed.
- In 1976, Appel and Haken gave the proof by computer:

K. Appel and W. Hakin, "Every planar map is 4-colorable," Bulletin of the AMS, Volume 82 (1976), 711-712.

Chapter 2. Existence problem

1. Introduction to existence problems
- 2. Basic proof methods**
3. Dirichlet principle (pigeonhole principle)



2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction



2.1. Direct proof

We begin with an example demonstrating the transitivity of divisibility.

Theorem. If a divides b and b divides c then a divides c .

Prove. By using the definition of the divisibility, there exist integers k_1 and k_2 such that

$$a = b k_1 \text{ and } b = c k_2.$$

Then

$$a = b k_2 = c k_1 k_2.$$

Let $k = k_1 k_2$. We have k as an integer, and $a = ck$, so by the definition of divisibility, a divides c .

2.1. Direct proof

If P, Then Q

- In most theorems, exercises or tests, you need to prove the form "If P, Then Q".
- In this example: "if a divides b and b divides c, then a divides c"
 - "P" is "If a divides b and b divides c" and "Q" is "a divides c".
- This is the standard state of many theorems.
- The direct proof can be conceived as a series of inferences beginning with "P" and ending with "Q":

$$P \Rightarrow \dots \Rightarrow Q$$

Most of proof is direct. When you have to prove, try starting with direct proof, unless you have a good reason not to.

2.1. Direct proof

- A **rational number** is a number that equals the quotient of two integers.
- Let Q denote the set of rational numbers.

Example: Proof the following theorem: **The sum of two rational numbers is rational.**

- Proof:
 - Let r and s be rational numbers.
 - Let $r = a/b$ and $s = c/d$, where a, b, c, d are integers, where $b, d > 0$
 - Then $r + s = (ad + bc)/bd$.
 - We know that $ad + bc$ is an integer.
 - We know that bd is an integer.
 - We also know that $bd \neq 0$.
 - Therefore, $r + s$ is a rational number.

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

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2.4. Proof by Mathematical Induction



2.2. Proof by contradiction

Requirement: Proof the statement P

Proof by contradiction:

- Assume it is false (Assume $\neg P$)
- Prove that $\neg P$ cannot occur
 - (it means a contradiction exists: not satisfying the properties given in the problem or come to the absurd such as $1 = 0$)

Requirement: Proof "If P , Then Q ",

Proof by contradiction:

- Assume it is false (Assume that " P and Not Q " are true).
- It thus means a contradiction exists.

2.2. Proof by contradiction

Example: Prove that if n is an integer and n^3+5 is odd, then n is even

Proof:

- $n^3+5 = 2k+1$ for some integer k (definition of odd numbers)
- $n^3 = 2k-4$
- $n = \sqrt[3]{2k-4}$

Umm, so direct proof didn't work out.

Rephrased: If n^3+5 is odd, then n is even

Proof by contradiction:

Assume that n^3+5 is odd, and n is odd are true

- n is odd $\rightarrow n=2k+1$ for some integer k (definition of odd numbers)
- $n^3+5 = (2k+1)^3+5 = 8k^3+12k^2+6k+6 = 2(4k^3+6k^2+3k+3)$ is odd ?!??
 - As $2(4k^3+6k^2+3k+3)$ is 2 times an integer, it must be even
 - Contradiction!

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction



2.3. Proof by contrapositive

Proof by contrapositive uses the logical equivalence of two statements "If P then Q" ($P \Rightarrow Q$) and "If not Q then not P" ($\neg Q \Rightarrow \neg P$):

$$(P \Rightarrow Q) \Leftrightarrow (\neg Q \Rightarrow \neg P)$$

Example:

The statement "If this is my car, then its color is red" is equivalent to

"If its color is not red, then it is not my car".

- Thus, to prove "If P, then Q" by using contrapositive proof, we prove "If not Q then not P".

2.3. Proof by contrapositive

Example 1. If x and y are two integer such that $x+y$ is even, then x and y are either both even or odd.

Proof. The contrapositive is

"If x and y are neither both even nor odd, then its sum is odd."

- As x and y are neither both even nor odd. Without loss of generality, we may assume x is even and y is odd. Then we could find integers k and m such that $x = 2k$ and $y = 2m+1$. Now we calculate $x+y = 2k + 2m + 1 = 2(k+m) + 1$, thus $x+y$ is odd.
- Therefore, we have proven the result by contraposition

2. Basic proof methods

2.1. Direct Proof

2.2. Proof by Contradiction

2.3. Proof by Contrapositive

2.4. Proof by Mathematical Induction



2.4. Proof by mathematical induction

- This is a very useful proof technique when we have to prove that the proposition $P(n)$ is true for all natural numbers $n \geq n_0$.
- Similar to the "domino effect" principle.

Outline of proof by Induction:

- Basic step: Prove the first statement $P(n_0)$ is true
- Inductive step: Given any integer $n \geq n_0$, prove that $P(n) \rightarrow P(n+1)$
(Assuming $P(n)$ is true and showing it forces $P(n+1)$ is true)
- Conclusion: $P(n)$ is true $\forall n \geq n_0$

(The assumption that $P(n)$ is true is called the **inductive hypothesis**)

2.4. Proof by mathematical induction

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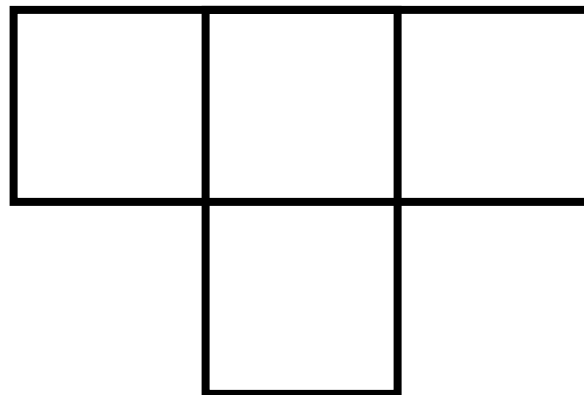
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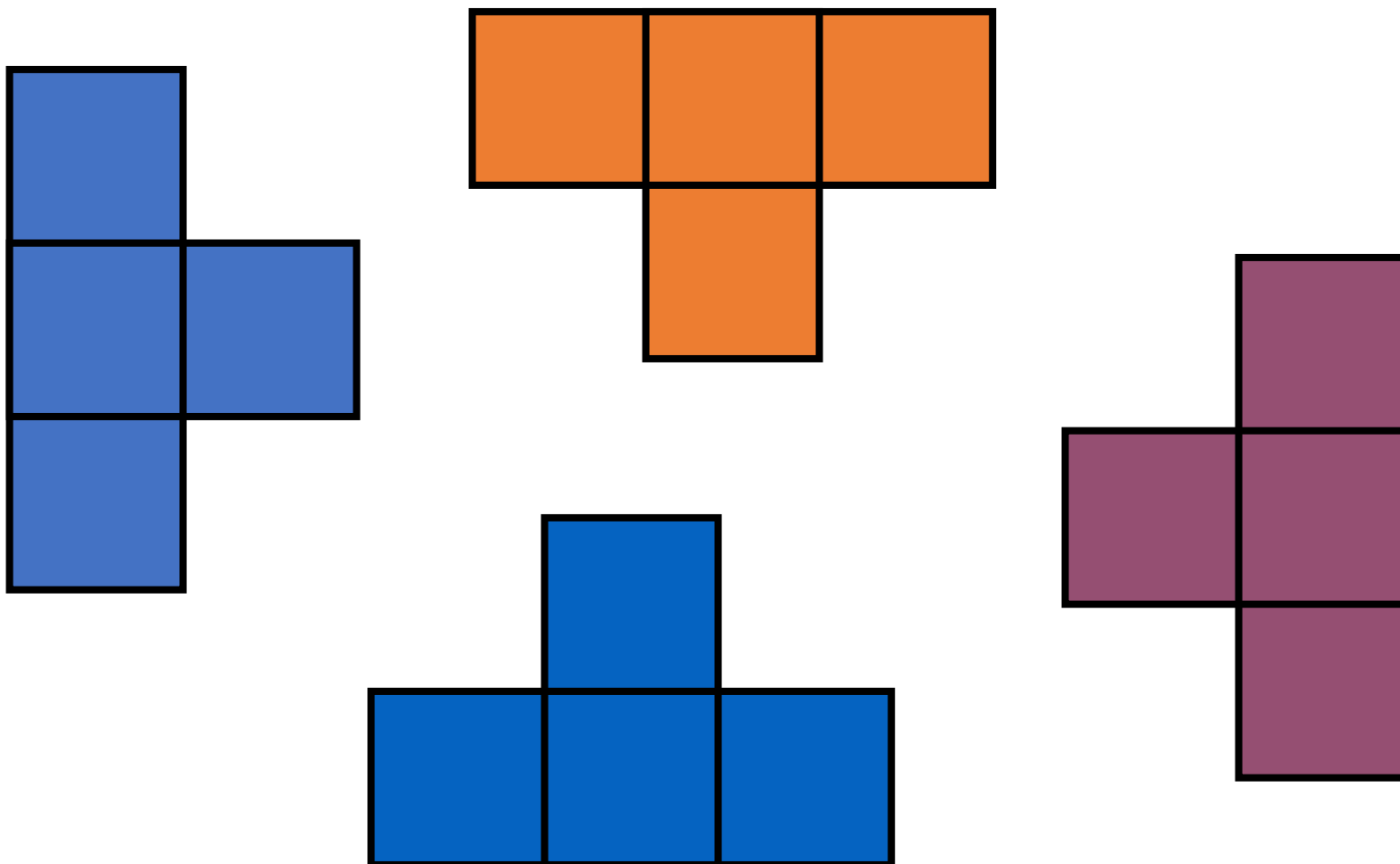
2.4. Proof by mathematical induction

Example: Prove that it is always possible to cover a chessboard of size $2^n \times 2^n$ ($n > 1$) with T-omino cards.



2.4. Proof by mathematical induction

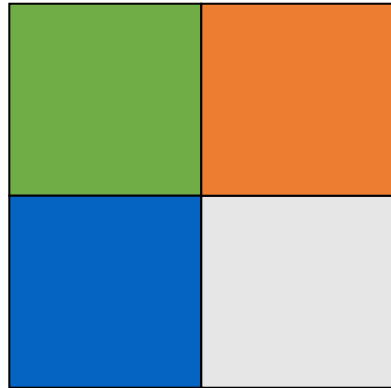
- Basic step: Prove that it is true when $n = 2$



2.4. Proof by mathematical induction

Assume we can cover chessboard of size $2^n \times 2^n$. We need to prove also can cover chessboard of size $2^{n+1} \times 2^{n+1}$.

Actually, divide the chessboard of size $2^{n+1} \times 2^{n+1}$ into 4 parts, each of size $2^n \times 2^n$. According to the **inductive hypothesis** (giả thiết qui nạp), each of these parts could be covered by T-ominos. Putting them on the chessboard of size $2^{n+1} \times 2^{n+1}$, we then get the answer for the problem.



Chapter 2. Existence problem

1. Introduction to existence problems
2. Basic proof methods
3. **Dirichlet principle (pigeonhole principle)**



3. Dirichlet principle (pigeonhole principle)

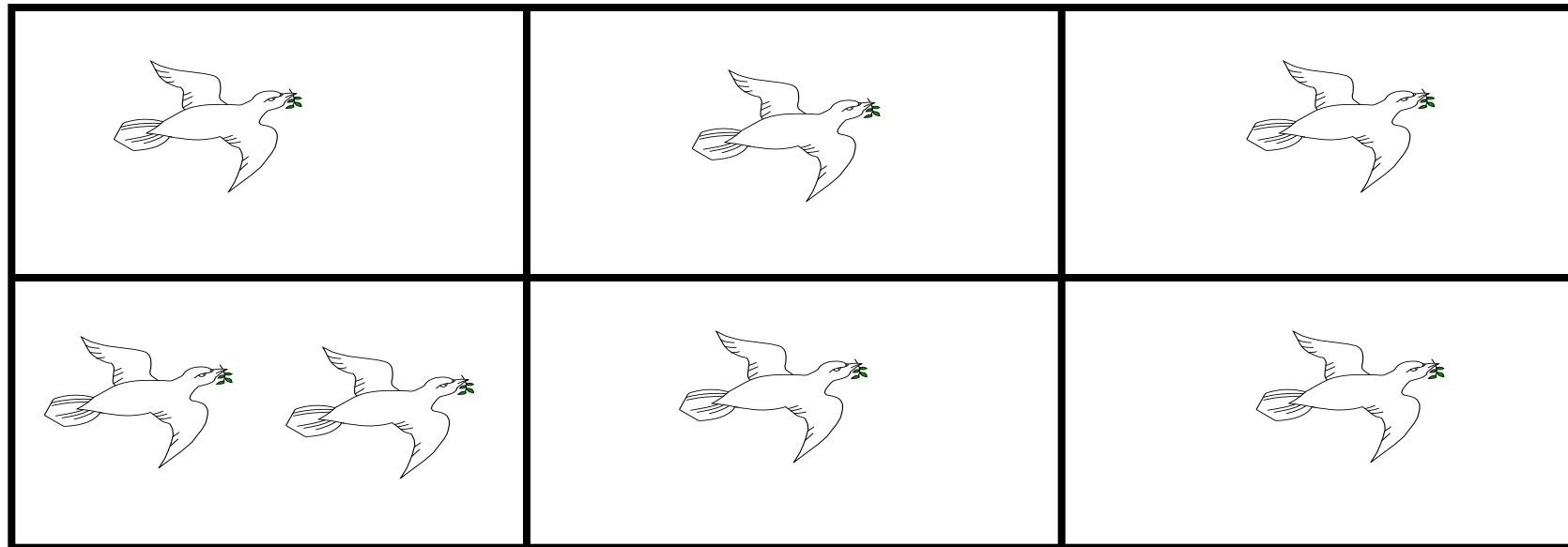
3.1. Principle statement

3.2. Application examples

3.1. Principle statement

If putting more than n objects into n boxes then at least one box has at least 2 objects (≥ 2).

- 7 objects
- 6 boxes



Proof. (Contradiction).

3.1. Principle statement

The above principle has been successfully applied by the German mathematician Dirichlet to solving many existence problems in combinatorics.

It is also presented in the language of pigeons:

"If one put more than n pigeons into n pigeonholes, then at least one hole has more than one pigeon (≥ 2)."

So the principle is also known as "**Pigeonhole principle**".

3.1. Principle statement

Example 1. Among 13 people, there are always 2 people born in the same month as there are only 12 months.

Example 2. In the exam, the test score is assessed by an integer between 0 and 100. Then at least how many students must take the test so that it is certainly to exist 2 students get same result ?

Solution. There are 101 different results

→ Using Dirichlet principle, the number of students is 102

3.1. Principle statement

Generalized Pigeonhole Principle: *If putting more than n objects into n boxes then at least one box has ≥ 2 objects.*

When the number of objects putting into k boxes is much larger than the k , it is obviously that the claim in the principle about the existence of a box containing at least two objects is too small. In such a case, we use the following generalized Dirichlet principle:

"If putting n objects into k boxes, one could always find at least one box containing $\geq \lceil n/k \rceil$ objects".

Here the symbol $\lceil \alpha \rceil$ is the least integer greater than or equal to α .

e.g.: $\lceil 3.14 \rceil = 4$

3.1. Principle statement

Dirichlet principle: "If putting n objects into k boxes, one could always find at least one box containing $\geq \lceil n/k \rceil$ objects".

Example 3. In a group of 100 people, what is the minimum number of people that were born in the same month ?.

Solution: Putting people born in the same month into one group. There are 12 months. Therefore, according to the Dirichlet principle, there exists at least one group consisting $\geq \lceil 100/12 \rceil = 9$ people

Example 4. There are 5 different types of scholarships. What is the minimum number of students so that there are at least 6 students getting the same type of scholarship?

Solution. The minimum number of students needed to ensure that 6 students getting the same scholarship is the smallest integer n such that $\lceil n/5 \rceil = 6$. The smallest integer is $n = 5 \times 5 + 1 = 26$.

Thus, 26 is the smallest number of student needed to ensure that there are 6 students getting the same type of scholarship

A large graphic on the left side of the slide. It features a dark blue background with a circular pattern of red dots of varying sizes, creating a sense of depth and movement. The word "HUST" is centered within this graphic in a white, bold, sans-serif font.

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THANK YOU !